



On the Role of MRI Sequences in Cross-Dataset Generalization for Brain Tumor Segmentation

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Problem and Motivation

Problem: brain tumor segmentation models lose accuracy on unseen datasets because of domain shift.

Motivation: few studies isolate how each MRI sequence contributes to robustness under cross-dataset shift.

- **RQ1:** how does segmentation performance degrade across domains as a function of the MRI sequence?
- **RQ2:** which MRI sequences are more robust in cross-dataset binary tumor segmentation?
- **RQ3:** how does target-domain annotation effort relate to performance gains for each modality?

Datasets

BraTS 2020: 369 pre-operative adult glioma cases from a limited number of institutions, four MRI sequences and masks.

BraTS 2024: 1,350 post-treatment cases from more sites and scanners, a more heterogeneous cohort.

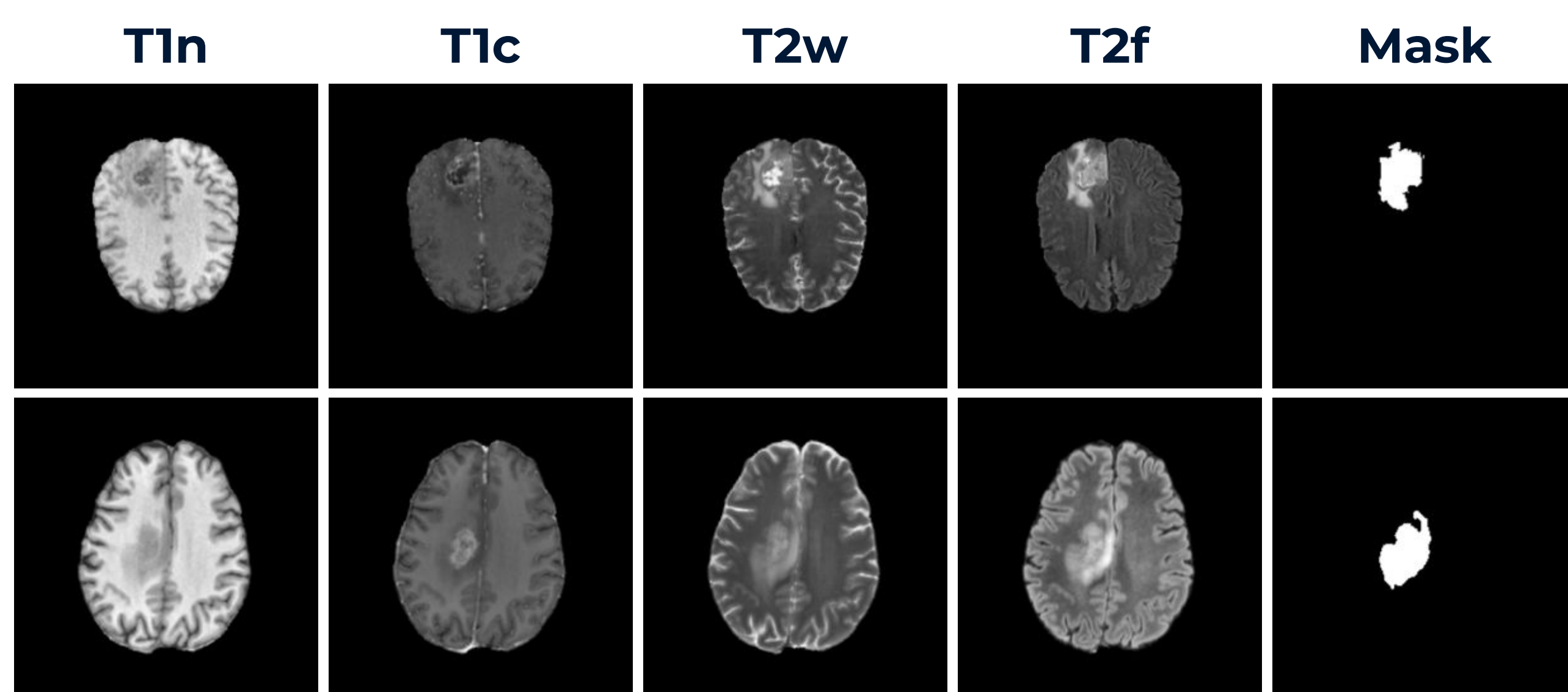


Figure 1. BraTS 2020 (top) and BraTS 2024 (bottom) examples across four MRI sequences and binary masks.

Method

2D ResUNet: ImageNet-pretrained ResNet-34 encoder and U-Net decoder with four up blocks and skip connections.

Input: a single-sequence 240×240 MRI slice.

Output: a binary tumor/non-tumor mask.

One model per sequence: 70/15/15 patient-level split, Dice + BCE loss, flips and rotations as data augmentation.

Results

Table 1. Dice / IoU (%) per protocol and sequence.

	T1c	T1n	T2w	T2f
Baseline 2020	61.3 / 51.2	62.9 / 52.9	73.3 / 64.2	77.3 / 68.4
Baseline 2024	66.1 / 54.7	67.3 / 56.0	74.2 / 63.5	78.3 / 68.6
Cross 20 → 24	49.7 / 38.8	49.7 / 39.1	62.3 / 51.4	58.2 / 46.9
Cross 24 → 20	65.3 / 55.5	58.3 / 48.8	70.4 / 61.2	76.2 / 67.6
Multi-domain	63.5 / 52.3	63.9 / 52.9	72.1 / 61.7	75.0 / 65.2

Results

Tumor size: evaluations per sequence across tumor-area quartiles, from Q1 (small) to Q4 (large).

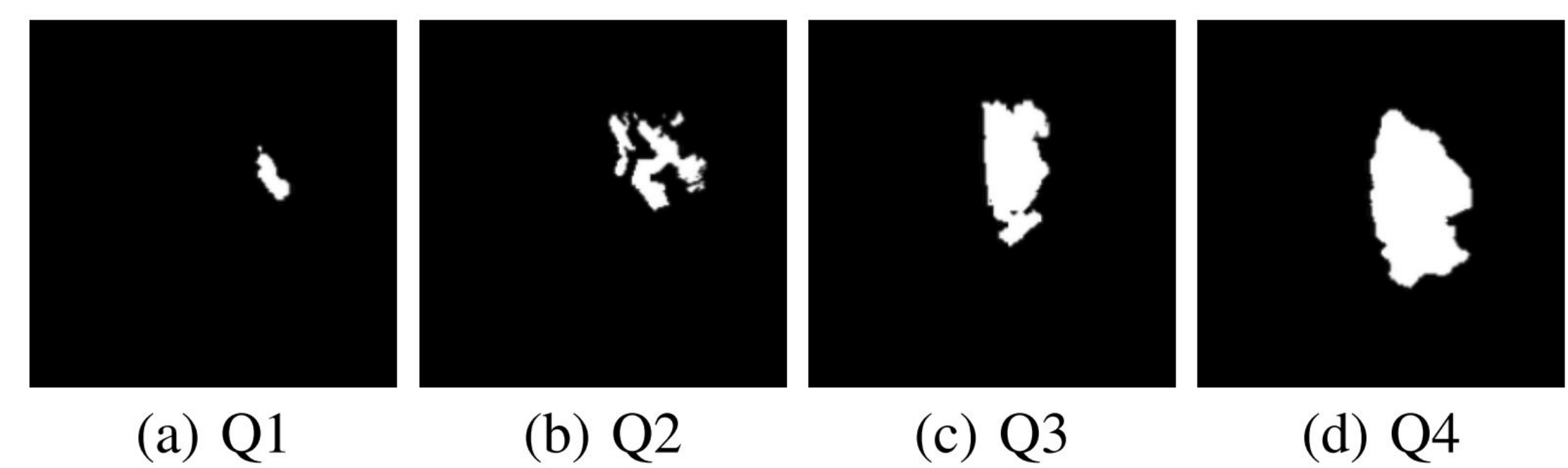


Figure 2. Representative masks for each tumor-area quartile.

Table 2. Dice / IoU (%) by tumor-size quartile; mean foreground area in the row label.

	T1c	T1n	T2w	T2f
Q1 · 171 px	26.9 / 19.6	26.0 / 18.8	39.5 / 30.3	49.0 / 39.2
Q2 · 729 px	63.3 / 51.1	64.0 / 51.4	75.3 / 63.8	77.6 / 66.8
Q3 · 1475 px	75.8 / 64.5	76.6 / 65.0	84.0 / 74.1	84.0 / 75.0
Q4 · 2957 px	79.7 / 68.6	81.3 / 70.1	85.7 / 76.4	84.8 / 76.0

Fine-tuning: FLAIR models adapted with labeled target data in 10% steps, in both directions.

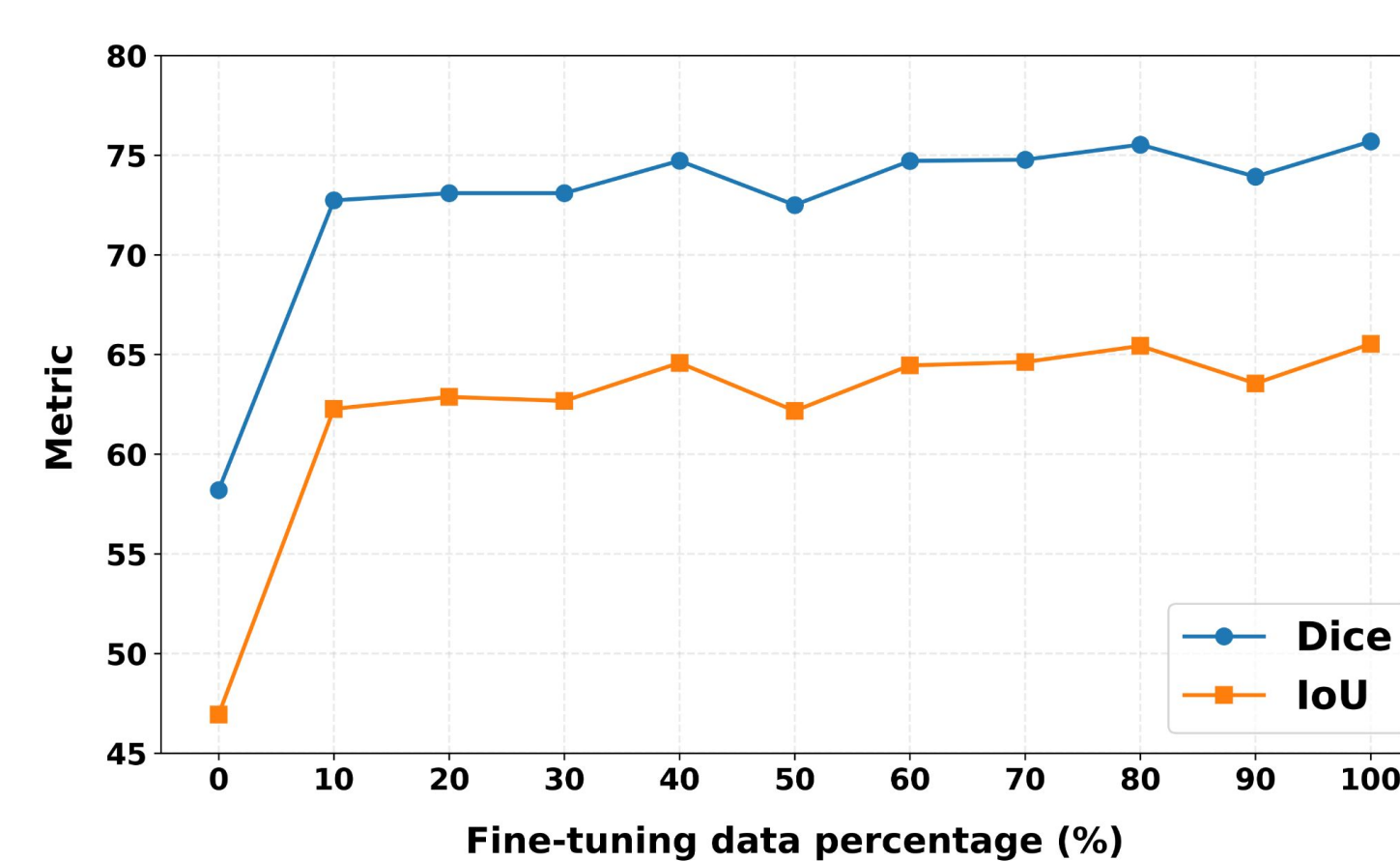


Figure 3. FLAIR, 2020 → 2024.

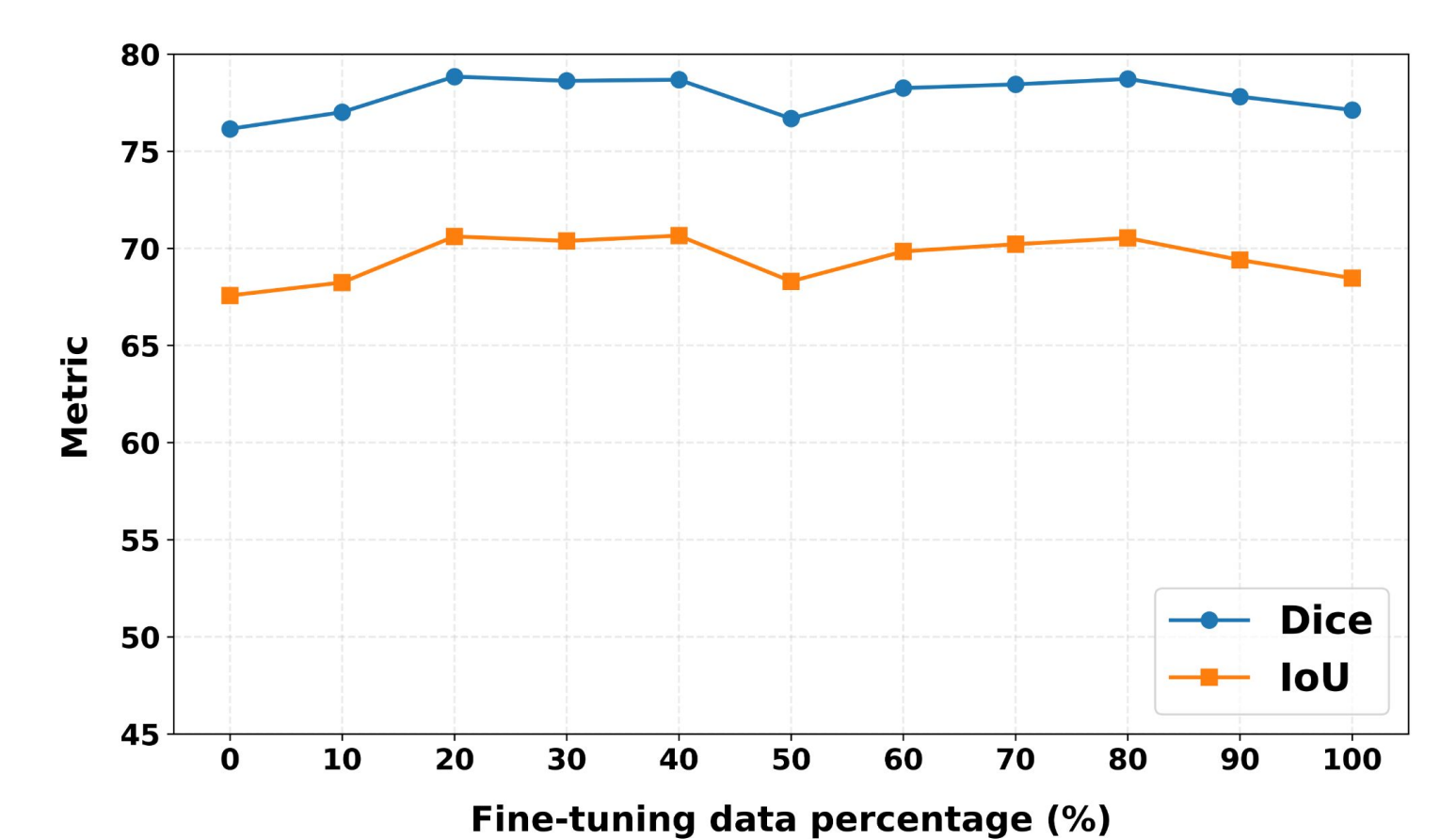


Figure 4. FLAIR, 2024 → 2020.

Qualitative Results

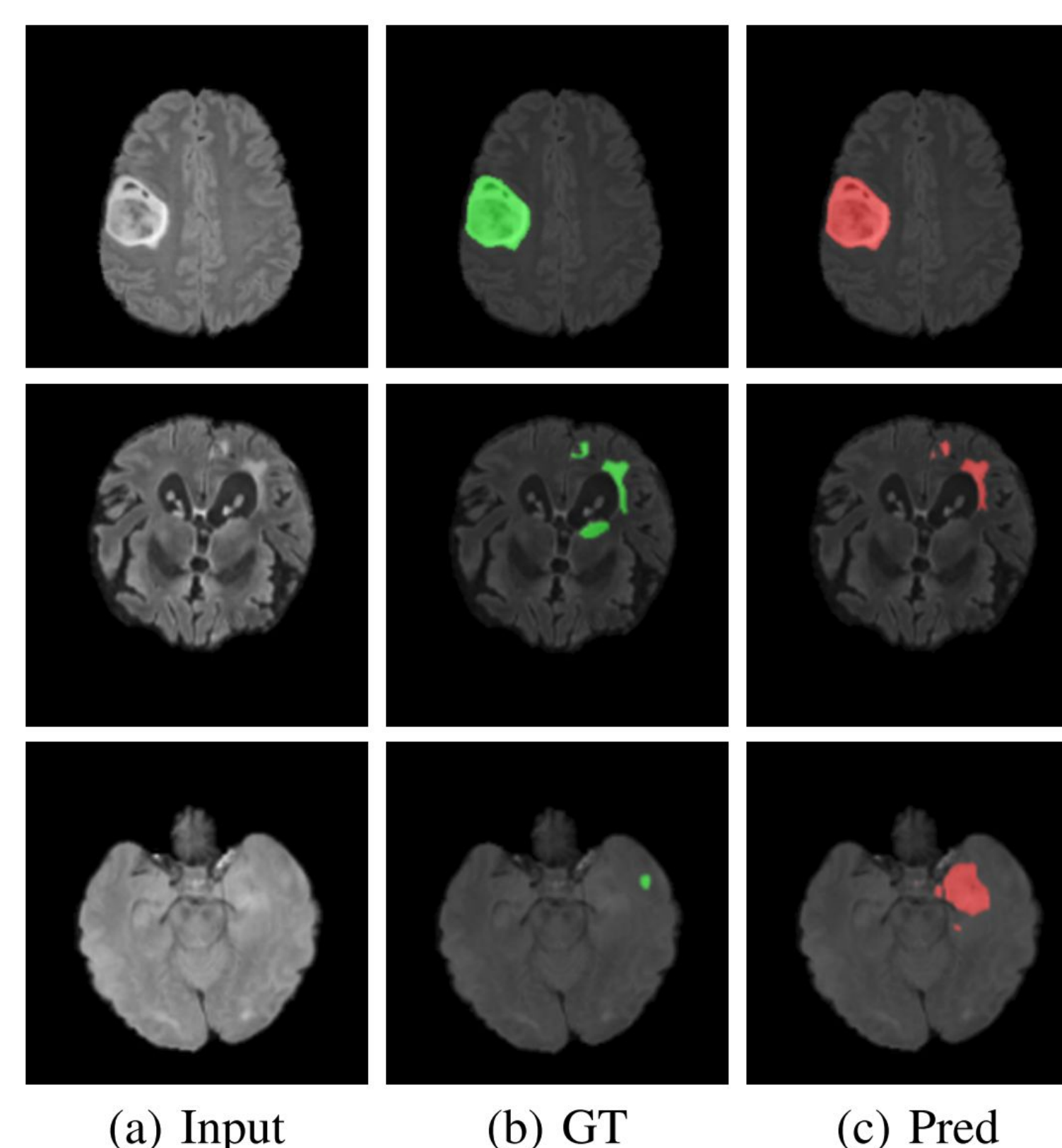


Figure 5. Segmentation examples: high-accuracy case (top, Dice: 99.1%), intermediate (middle, Dice: 70.1%), and failure case (bottom, Dice: 0.0%).

Conclusions

T2 sequences, mainly **T2f/FLAIR**, transfer best across datasets.

Models trained on BraTS 2024 transfer better to BraTS 2020 than vice versa.

Multi-domain training improves robustness: 75.0 Dice with FLAIR.

Limited target-domain annotation already provides substantial gains.

Modality choice, data diversity and lesion size drive results.

Acknowledgments

